Week of August 24, 2026:
Nesso-1 and Boltz-2 on Runs N' Poses

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Repository: `BindingAffinityPredictor/weeks/2026-W35`

Executive summary

This week I focused on four questions:

1. I measured the speed of Nesso-1 and Boltz-2 on the same local hardware and input cohort.
2. I used Runs N' Poses to compare Nesso-1 and Boltz-2 on the same 50 protein–ligand systems. I selected the cohort using Boltz-2's June 1, 2023 structural-training cutoff.
3. I reviewed the principal benchmarks used by the FEP community, including what data they contain, where the measurements came from, and what each benchmark tests.
4. I reviewed the Nesso-1 architecture, from protein and ligand representations through pair features, distance prediction, pocket selection, and affinity prediction.

1 Measured speed comparison

I timed both models on the new 50-system cohort using the same 10-GiB NVIDIA RTX 3080, warm local checkpoint caches, one worker, and seed 42. Nesso-1 used five recycling steps. Boltz-2 used three recycling steps, one 200-step diffusion sample, and an MSA subsample of 1,024 sequences.

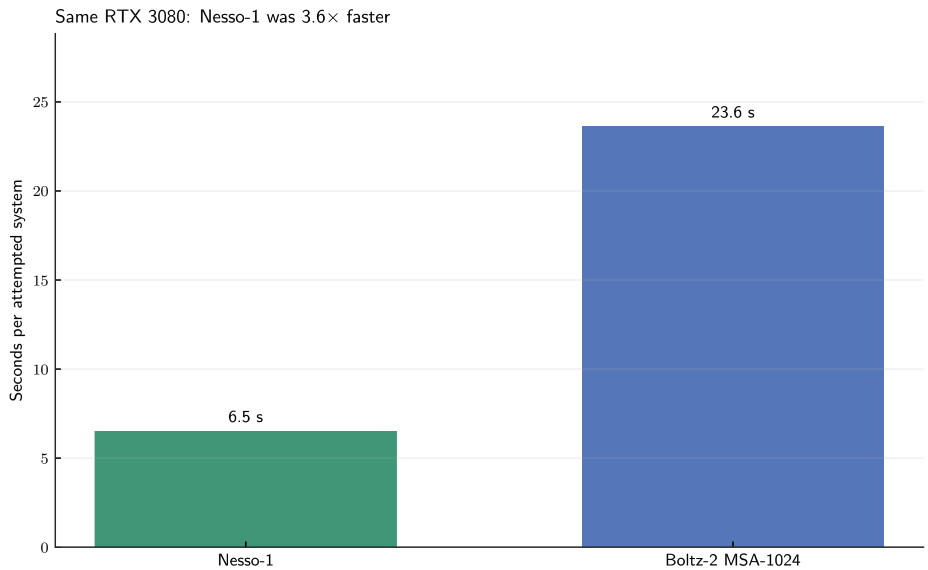

Figure 1: Measured local preprocessing and inference time divided by the 50 attempted systems. The external ColabFold MSA search is excluded.

Workflow	Completed	Total wall time	Seconds/attempt
Nesso-1	50/50	326.1 s	6.52
Boltz-2 MSA-1024	49/50	1181.7 s	23.63

Table 1: Nesso-1 was 3.62 times faster in this specific local workflow.

This is not an equal-output kernel benchmark. Boltz-2 generated full Cartesian protein–ligand complexes, whereas Nesso-1 stopped at a residue-center and ligand-heavy-atom distance representation. Boltz-2's external MSA search time is also absent. One 818-residue Boltz-2 system exceeded the available GPU memory; the failure remains in the completion count.

2 Paired Runs N' Poses experiment and results

Runs N' Poses contains 2,600 high-resolution protein–ligand systems released after September 30, 2021 [1]. It provides the experimental complex structures and precomputed similarities to older protein–ligand structures. I selected 50 systems released after June 1, 2023, with ten systems in each familiarity interval 0–20, 20–40, 40–60, 60–80, and 80–100.

2.1 Model inputs

For every system, both models received

$$\text{exact deposited protein sequence} + \text{stereochemical ligand SMILES.}$$

Boltz-2 additionally received a ColabFold MSA derived from the protein sequence.

Nesso-1 completed 50/50 systems. Boltz-2 completed 49/50; the 818-residue 8UK6 system exceeded the 10-GiB GPU limit under the frozen MSA-1024 protocol. Direct model-to-model comparisons therefore use the **same 49 complexes** completed by both models. The remaining Nesso-1-only result is retained in its completion record but excluded from paired accuracy comparisons.

2.2 What familiarity means

For test complex i and cutoff date c, familiarity is schematically

$$F_i(c) = \max_{j:\,\mathrm{date}(j)<c} \mathrm{similarity}(i, j).$$

The published `sucos_shape_pocket_qcov` score ranges from 0 to 100 and combines ligand-pose similarity with coverage of the experimental protein pocket. A high score means that a closely related pocket–ligand pose existed before the cutoff; it is not a model prediction and not an accuracy score.

Here, "cutoff" means the **PDB structural-training cutoff**. Nesso-1's structural trunk used experimental PDB structures released before September 30, 2021; Boltz-2 used experimental PDB structures released before June 1, 2023 [2, 3]. Consequently, the same physical complex has two potentially different familiarity scores. **Only the historical PDB lookup changes; the test complex itself does not.** The two model columns below are deliberately kept separate; their x-values should not be treated as one shared measurement.

2.3 Shared structural metrics

The closest common representation is a distance between a protein Cβ token (Cα for glycine) and a ligand heavy atom. Nesso-1 supplies an expected distance from its 64-bin distribution; Boltz-2 supplies the distance measured in its sampled coordinate model. The two F1 metrics classify different objects: token-contact F1 classifies individual residue-token–ligand-atom pairs, whereas pocket F1 classifies whole protein residues.

Metric	Plain-language meaning	Better
Interface distance MAE	Mean absolute error for token pairs whose experimental distance is at most 15 Å.	Lower
Token-contact F1	Recovery of individual protein-token–ligand-atom contacts within 6 Å, balancing precision and recall.	Higher
Pocket F1	Recovery of the experimental 6 Å pocket as a set of protein residues.	Higher

Table 2: The three metrics applied to both model outputs. Exact ligand-graph symmetries were handled before scoring.

For token-contact F1, an experimental positive is a Cβ–ligand-heavy-atom distance at most 6 Å (Cα for glycine). Nesso-1 predicts that pair as a contact when the expected distance from its distogram is at most 6 Å; Boltz-2 is thresholded using the corresponding distance in its sampled coordinate model. For pocket F1, the experimental pocket contains a residue when *any* of its resolved heavy atoms lies within 6 Å of any ligand heavy atom. Boltz-2 can apply the same all-heavy-atom definition to its generated coordinates. Nesso-1 has no generated side-chain coordinates, so I call a residue a predicted pocket residue when its Cβ/Cα token has an expected distance of at most 6 Å to at least one ligand atom. Nesso-1's pocket F1 is therefore a coarse residue-center approximation to an atomic pocket, not a coordinate-based pocket measurement. I did not use Nesso-1's native pocket mask here: its 15 Å affinity cutoff serves model context and is not the benchmark's 6 Å contact definition. Experimental coordinates were introduced only during this post-inference scoring.

2.4 Accuracy as a function of familiarity

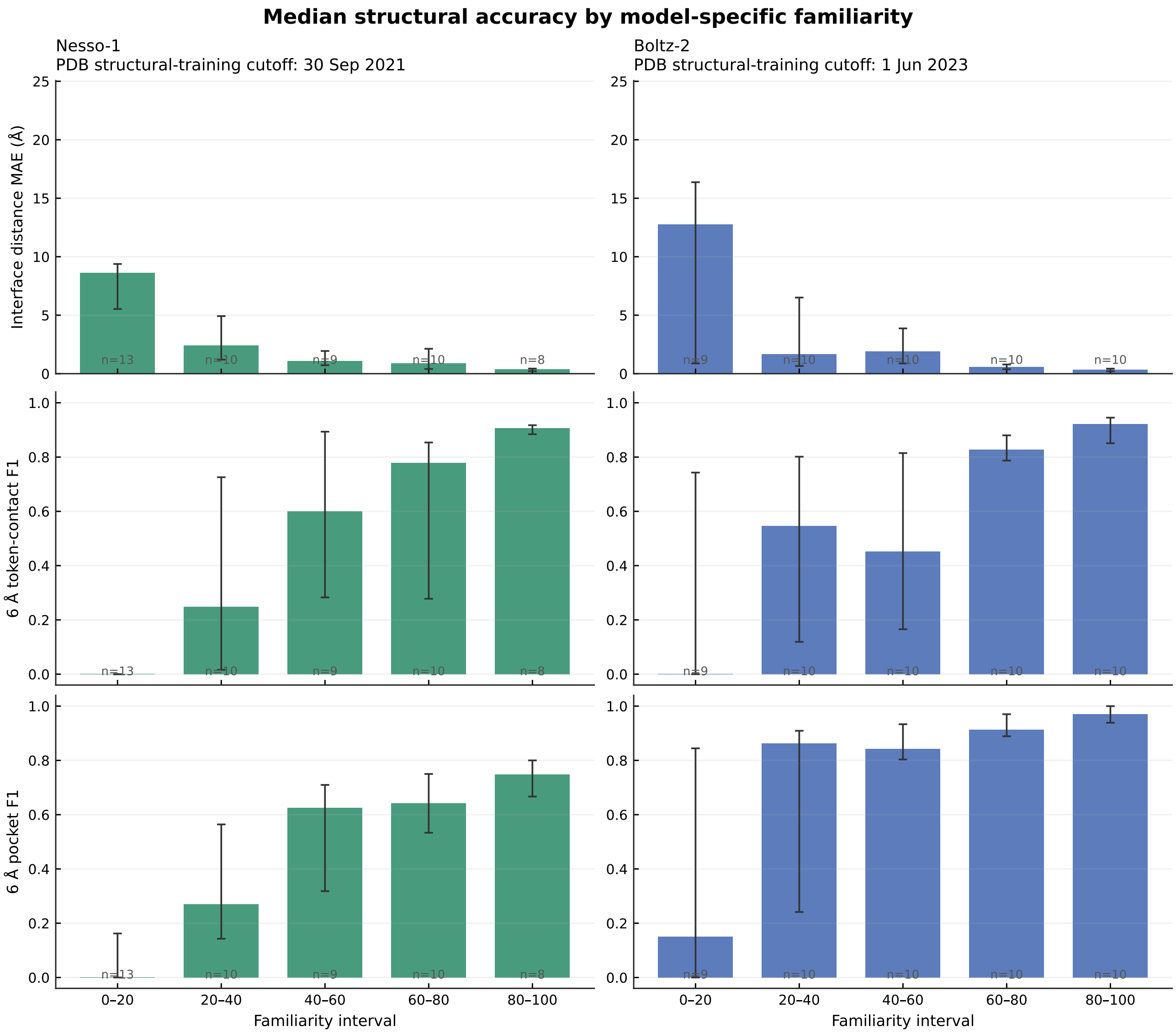

Figure 2: Structural accuracy versus model-specific familiarity. Each bar is the median in a 20-point familiarity interval; error bars are 95% bootstrap intervals, and n is the number of systems in that interval. Lower interface MAE and higher F1 are favorable.

Model and PDB structural-training cutoff	Interface MAE ρ	Contact F1 ρ	Pocket F1 ρ
Nesso-1, Sep. 2021 ($n = 50$)	-0.721 $[-0.876, -0.486]$	$+0.727$ $[+0.511, +0.876]$	$+0.677$ $[+0.433, +0.854]$
Boltz-2, Jun. 2023 ($n = 49$)	-0.690 $[-0.839, -0.475]$	$+0.692$ $[+0.485, +0.828]$	$+0.690$ $[+0.472, +0.831]$

Table 3: Spearman correlations with 95% bootstrap intervals. All six intervals exclude zero.

Both models show the same high-level behavior: greater pre-cutoff structural familiarity is associated with lower interface-distance error and better contact and pocket recovery. The bin medians fluctuate, especially in low-familiarity bins; the continuous rank correlations are the primary evidence for a broad trend.

Nesso-1 improves relatively gradually across its five bins. Boltz-2's pocket recovery is already high in the 20–40 bin, but its lowest-familiarity bin contains a mixture of successes and complete failures, giving very wide uncertainty intervals. Its contact F1 and interface MAE also fluctuate between the 20–40 and 40–60 bins before improving at higher familiarity. These descriptive bin differences should not be interpreted as a head-to-head comparison at the same x-value because the two columns use different historical cutoffs and the cohort was balanced only by Boltz-2's score.

This association does not establish its cause. Familiarity may reflect direct training-set resemblance, but protein family, ligand class, experimental structure quality, and intrinsic problem difficulty can also vary with the score. The result therefore does not by itself prove memorization.

2.5 Direct comparison on the same systems

Metric	Nesso-1 median	Boltz-2 median	Boltz/Nesso/ties
Interface distance MAE (Å)	1.387	0.881	22/27/0
6 Å token-contact F1	0.462	0.780	27/16/6
6 Å pocket F1	0.514	0.889	44/1/4

Table 4: Paired results on the 49 systems completed by both models. The final column counts which model was better on each individual system.

Boltz-2 had substantially better pocket recovery: its mean paired improvement in pocket F1 was +0.318 with a 95% bootstrap interval of [+0.251, +0.388]. Contact F1 also favored Boltz-2 on average. Interface MAE was mixed: Nesso-1 was better on 27 systems, Boltz-2 on 22, and the paired mean improvement interval included zero. The lower Boltz-2 cohort median should therefore not be read as a consistent system-by-system MAE advantage.

Only Boltz-2 generates explicit coordinates. After global protein Cα alignment, its median protein Cα RMSD was 0.894 Å and its median symmetry-corrected ligand heavy-atom RMSD was 3.355 Å. Ligand RMSD was below 2 Å for 22/49 structures and below 5 Å for 30/49. Nesso-1 has no valid coordinate RMSD because it does not output a Cartesian complex.

3 Benchmarks used by the FEP community

Most free-energy perturbation (FEP) benchmarks ask the following question: given a prepared protein structure, a plausible bound pose, and a series of related ligands, can a method recover the experimental *relative* affinity differences within that series? In Table 5, *data vintage* therefore means the newest date that can be audited from the public record: the publication year of the affinity measurement where available, or the challenge/data-release date for blinded pharmaceutical sets. Links in that column point to the underlying paper, metadata, or public data release rather than merely to the benchmark paper.

Benchmark	Data vintage and provenance	Approximate scale	Main question	Paper / data
Original FEP+/JACS set	Public assay papers span roughly 2004–2013; the collection was assembled in 2015.	199 ligands, eight protein systems	Can relative binding free energies rank congeneric ligand series?	Paper; data
FEP+4 used in Week 34	Same source pool; the newest clearly linked assay series is TYK2, 2013.	87 ligands, four kinase series	Can related ligands be ranked within CDK2, JNK1, p38, and TYK2?	TYK2 source; data
D3R Grand Challenges	The latest round, GC4, used blinded Janssen and Novartis data in 2018–2019.	Challenge-dependent, blind test sets	Can methods predict poses, affinity ranks, and in some rounds relative free energies prospectively?	GC4 paper; data
Hahn/OpenFF benchmark	Many source papers; each ligand records a measurement DOI. The latest source year in the released metadata is 2019.	599 ligands, 22 systems in the published set	Can free-energy methods be compared on open, standardized inputs and measurements?	Paper; repository
Ross FEP benchmark	Prior source/FEP studies extend through 2022; 93 compounds are from anonymized in-house sets.	1,237 compounds across many series	How does FEP perform across a broader domain, including charge changes, fragments, macrocycles, scaffold hops, GPCRs, and buried-water changes?	Paper; repository
OpenFE public study	No newer compound set: it explicitly reuses the Ross v2.0 collection.	JACS, fragment, and Merck subsets of the Ross collection	Can an open RBFE workflow reproduce results on shared inputs and protocols?	Protocol; repository
CASP16 affinity task	Blinded pharmaceutical data in the 2024 challenge; public experimental-data release dated 7 August 2025.	122 assessed ligand cases across two affinity targets	In a blind challenge, can methods rank affinities before and after experimental poses are released?	Paper; data

Table 5: The benchmarks were built for different scientific purposes and should not be interpreted as interchangeable leaderboards [4–9].

4 Nesso-1 architecture, step by step

The central architectural idea of Nesso-1 is:

Protein sequence + ligand chemistry $\longrightarrow$ token features $\longrightarrow$ pair representation z_{ij} $\longrightarrow$ distance distributions $\longrightarrow$ predicted pocket $\longrightarrow$ affinity outputs.

Nesso-1 does not first generate a complete atomic protein–ligand complex and then score that structure. Instead, it learns a probabilistic map of distances between protein-residue centers and ligand heavy atoms, then gives this learned geometry to a smaller affinity network [2].

4.1 Inputs and tokenization

For the simplest use case, the input contains a protein amino-acid sequence, a ligand SMILES string, and the identity of the ligand to score. Nesso-1 does not require an experimental protein structure, an MSA, a user-specified pocket, or an initial protein–ligand pose.

Each standard protein residue becomes one token. Its geometric target is the Cβ atom, or Cα for glycine. Each ligand heavy atom becomes one token; ligand hydrogens are omitted. If the protein has N_P residues and the ligand has N_L heavy atoms, the model begins with approximately

$$N = N_P + N_L$$

tokens. For example, a 300-residue protein with a 30-heavy-atom ligand gives approximately 330 tokens.

The input-processing code constructs standard reference atoms for each amino acid. For a ligand supplied as SMILES, it generates an isolated 3D conformer with RDKit ETKDG. This conformer provides the ligand's internal geometry; it does *not* specify where the ligand sits relative to the protein. The atom encoder uses atom identity, names, charge, chirality, hybridization, bonds, and local reference geometry, then pools atom information into token vectors.

4.2 ESM-2 supplies protein sequence context

The protein sequence is also processed by the pretrained ESM-2 protein language model. For each residue i, ESM-2 supplies a contextual vector

$$\mathbf{e}_i \in \mathbb{R}^{1280}.$$

"Contextual" means that the vector for an amino acid depends on the surrounding sequence, not only on its residue identity. Nesso-1 uses ESM-2 in place of the explicit MSA processing used by Boltz-2.

After the atom, residue-type, and ESM information is projected, every token has a single-token representation $\mathbf{s}_i$. In the released Nesso-1 checkpoint,

$$\mathbf{s}_i \in \mathbb{R}^{384}.$$

The 384-dimensional vector does not replace ESM-2 before ESM is used. Nesso first applies a learned projection to each 1280-dimensional ESM vector,

$$\widetilde{\mathbf{e}}_i = W_2 \ \mathrm{ReLU}(W_1 \ \mathrm{LayerNorm}(\mathbf{e}_i)) , \qquad \mathbb{R}^{1280} \longrightarrow \mathbb{R}^{384}.$$

It then adds the projected ESM context to Nesso's atom/residue token features,

$$\mathbf{s}_i = \mathbf{s}_i^{\text{atom/residue}} + \widetilde{\mathbf{e}}_i^{\text{ESM}}, \qquad \mathbf{s}_i \in \mathbb{R}^{384}.$$

The projection is learned: it selects and transforms the parts of ESM-2 that are useful to Nesso rather than selecting the first 384 ESM coordinates.

For a 300-residue protein and a 30-heavy-atom ligand, the shapes can be followed as

$$\underbrace{300 \times 1280}_{\text{raw protein ESM-2}} \quad \longrightarrow \quad \underbrace{300 \times 384}_{\text{projected protein context}} \quad ,$$

followed, after alignment with the ligand tokens, by

$$\underbrace{330 \times 384}_{\text{all token features}} \ \longrightarrow \underbrace{330 \times 330 \times 128}_{\text{pair representation } z}.$$

Thus 1280, 384, and 128 describe three different objects: the raw ESM residue embedding, Nesso's combined single-token feature, and each entry in the token-pair matrix, respectively. The compression is especially important for the pair representation because its memory already grows as N^2; using width 1280 instead of 128 there would require approximately ten times the pair-feature memory.

4.3 The pair representation z

The central object is not a coordinate array but a learned pair tensor

$$z \in \mathbb{R}^{N \times N \times 128}.$$

Each element z_{ij} is a 128-dimensional description of the relationship between tokens i and j. The matrix contains protein–protein, protein–ligand, ligand–protein, and ligand–ligand blocks. For example, z_{42,L_5} represents what the network currently knows about protein residue 42 relative to ligand atom 5.

The initial pair representation is built approximately as

$$z_{ij}^{(0)} = W_{\mathrm{L}}\mathbf{s}_i + W_{\mathrm{R}}\mathbf{s}_j + \mathbf{r}_{ij} + \mathbf{b}_{ij}.$$

Here W_{L} and W_{R} are learned projections from 384 to 128 dimensions, $\mathbf{r}_{ij}$ describes relationships such as sequence separation and chain or entity identity, and $\mathbf{b}_{ij}$ describes ligand bonds and bond types. Projected ESM information supplies an additional single-to-pair update. Thus, $z_{ij}^{(0)}$ is an addition of learned token and pair features; it is not the scalar dot product $\mathbf{s}_i^{\mathsf{T}}\mathbf{s}_j$.

At initialization, $z_{ij}^{(0)}$ mainly records what kinds of objects i and j are and their known sequence or bond relationship. Its useful geometric meaning is developed by the Pairformer.

4.4 What the Pairformer does

The released trunk contains 48 Pairformer blocks. Every block applies, in order:

1. outgoing triangle multiplication;
2. incoming triangle multiplication;
3. starting-node triangle attention;
4. ending-node triangle attention; and
5. a pair-transition multilayer perceptron.

Each operation produces a residual update, $z \leftarrow z + \Delta z$, so the network progressively refines rather than replaces the pair table.

Triangle multiplication. To update pair (i, j), triangle multiplication aggregates information through every possible third token k. Its two directions can be written schematically as

$$\Delta z_{ij}^{\mathrm{out}} \sim \sum_k A(z_{ik}) \odot B(z_{jk}), \qquad \Delta z_{ij}^{\mathrm{in}} \sim \sum_k A(z_{ki}) \odot B(z_{kj}),$$

where A and B are learned projections and $\odot$ denotes elementwise multiplication. These equations show the information flow rather than every normalization and gate in the implementation.

For example, let i be protein residue 42, j ligand atom 5, and k ligand atom 6. If atoms 5 and 6 are bonded and atom 6 is predicted near residue 42, the two other sides of the triangle provide information about the possible relationship between residue 42 and atom 5.

Triangle attention. Attention also considers third tokens k, but learns which intermediates are most relevant. One attention direction is schematically

$$\alpha_{ijk} = \mathrm{softmax}_k\left(\frac{\mathbf{q}_{ij}^{\mathsf{T}}\mathbf{k}_{ik}}{\sqrt{d}} + \text{pair bias}\right), \qquad \Delta z_{ij} = \sum_k \alpha_{ijk}\mathbf{v}_{ik}.$$

The released blocks use four attention heads, allowing different heads to emphasize different relational patterns. Dot products therefore occur *inside* attention when calculating weights, but the complete z tensor is not a dot-product similarity matrix.

Pair transition. After communication among pairs, a small MLP transforms the 128 features within each pair independently,

$$z_{ij} \leftarrow z_{ij} + \mathrm{MLP}(z_{ij}).$$

Triangle operations communicate across the pair table; the transition mixes features within one table entry.

The Pairformer does not analytically enforce molecular geometry. Instead, the distogram training loss propagates through all 48 blocks, rewarding internal features that predict experimental distance bins. This supervision is what makes the final z geometrically informative. Before the distogram head, the released implementation symmetrizes it as

$$z_{ij}^{\mathrm{sym}} = z_{ij} + z_{ji},$$

consistent with the physical symmetry $d_{ij} = d_{ji}$.

Unlike a full AlphaFold-style trunk, the Nesso-1 trunk updates only z. It does not maintain a large evolving MSA representation and it does not send z into an atomic diffusion decoder.

4.5 From z to a distogram

A linear head converts every pair representation into 64 logits. Softmax gives a categorical distance distribution,

$$p_{ij}^{(b)} = P(d_{ij} \text{ lies in distance bin } b).$$

Thus, the model can assign probability to several possible distances instead of committing to one coordinate set. Using bin center c_b, its expected distance is

$$\bar{d}_{ij} = \sum_b p_{ij}^{(b)} c_b.$$

The uncertainty of a pair distribution can be summarized by

$$H_{ij} = -\sum_b p_{ij}^{(b)} \log p_{ij}^{(b)}.$$

Nesso reports averaged protein–protein, protein–ligand, and ligand–ligand entropies. A concentrated distance distribution has lower entropy; a diffuse one has higher entropy.

This distogram is Nesso-1's native structural output. The paper reconstructs approximate residue-center and ligand-heavy-atom poses by optimizing coordinates to agree with the expected distances, but the standard released inference path does not generate a full atomic protein PDB structure.

4.6 Automatic pocket selection and recycling

The model infers the relevant protein region rather than receiving a pocket from the user:

1. The first trunk pass processes the complete protein and ligand.
2. The distogram head estimates each protein residue's distance to the ligand.
3. Protein tokens predicted within 22 Å of the ligand are retained, subject to a default total crop budget of 256 tokens.
4. Later recycling passes repeatedly update z using this smaller region.

Our command used `recycling_steps=5`. In the released implementation, this means one initial trunk evaluation followed by five recycled evaluations. For the affinity calculation, a tighter predicted cutoff of 15 Å selects the protein tokens supplied to the affinity modules. These cutoffs define model context; they should not be interpreted as exact physical boundaries of the binding site.

4.7 The affinity modules

The selected interface is passed to a second, smaller Pairformer. It receives:

- token features from the atom and residue encoders;
- the ESM-2 protein embeddings;
- the pretrained pair representation z; and
- the predicted distance-bin probabilities.

The released checkpoint contains two independently parameterized affinity modules, each with eight Pairformer blocks. Their masks focus computation on protein–ligand and ligand–ligand pairs. After these updates, the relevant pair features are averaged into one global vector,

$$\mathbf{g} = \frac{\sum_{ij} m_{ij} z_{ij}}{\sum_{ij} m_{ij}},$$

and small multilayer perceptrons map $\mathbf{g}$ to a continuous affinity output and a binary binder output.

The two continuous module outputs are

$$\texttt{affinity\_pred\_value1} \qquad \texttt{affinity\_pred\_value2}.$$

The value normally used is their mean,

$$\texttt{affinity\_pred\_value} = \frac{\texttt{value1} + \texttt{value2}}{2}.$$

The two binder probabilities are likewise averaged. The released documentation expresses the continuous value as

$$\log_{10}\left(\frac{\mathrm{IC}_{50}}{\mu\mathrm{M}}\right),$$

so lower values indicate stronger predicted binding: -3, 0, and 2 correspond nominally to approximately 1 nM, 1 μM, and 100 μM. Because the training labels mix several assay endpoints, this output is best treated as an assay-trained potency score, not as a rigorous calculated IC_{50} or binding free energy.

The binary output is

$$P(\text{binder} \mid \text{protein, ligand}),$$

which addresses classification rather than continuous potency ranking. The two outputs should remain separate in evaluation.

4.8 Training objectives

Training is most easily understood in two stages. First, the atom encoder and trunk learn geometry by minimizing categorical cross-entropy for experimental distance bins. Structural training uses PDB structures released before September 30, 2021, AFDB protein distillation data, and distilled protein–ligand complexes described in the paper. This teaches z to encode pairwise geometry.

Second, the affinity modules learn public assay outcomes. The regression data come mainly from BindingDB and ChEMBL, while classification data also include PubChem, the CeMM fragment collection, and MIDAS. Classification uses focal loss. Regression uses Huber losses for both absolute values and within-assay differences. The relative term encourages

$$\hat{y}_A - \hat{y}_B \approx y_A - y_B$$

for compounds measured in the same assay. It is up-weighted because public assays can have different systematic offsets, while within-assay ligand ordering is often the more reliable medicinal-chemistry signal.

Affinity and potency labels K_d, K_i, IC_{50}, and EC_{50} are treated interchangeably during this training. This choice expands the public training set but limits the strict physical interpretation of the continuous output.

4.9 Bridge to Boltz-2

Component	Nesso-1	Boltz-2
Protein context	ESM-2 sequence embedding; no MSA required	MSA-derived evolutionary information, with optional templates
Central internal object	Token-pair representation and distance distributions	Single and pair representations coupled to full coordinate generation
Protein geometric resolution	One predicted center per residue	Full atomic complex output
Ligand geometric resolution	Heavy-atom tokens	Full atomic complex output
Structure decoder	No atomic diffusion; optional distance-based reconstruction	Diffusion decoder generates explicit coordinates
Pocket handling	Inferred from expected protein–ligand distances	Emerges through complex cofolding
Affinity calculation	Two smaller Pairformer modules operating on the predicted interface	Affinity head operating on Boltz trunk and structural representations
Primary design priority	Fast affinity classification and ranking	Detailed complex structure together with affinity

Table 6: High-level architectural bridge from Nesso-1 to Boltz-2. A detailed Boltz-2 forward pass is the next architecture topic.

Nesso-1 therefore saves computation by stopping at a probabilistic residue-center/heavy-atom distance map. The tradeoff is that it does not directly resolve protein side-chain conformations, precise hydrogen-bond angles, atomic steric clashes, explicit waters, or a complete Cartesian complex. Its hypothesis is that the learned interface map is nevertheless sufficient for rapid affinity ranking.

5 Conclusion and limits

The paired experiment gives two main results. First, structural familiarity is strongly associated with accuracy for both Nesso-1 and Boltz-2, using a model-appropriate historical cutoff for each. Second, Boltz-2's explicit coordinate generation recovers the experimental pocket more reliably, whereas Nesso-1 is faster and remains competitive on the common interface-distance error metric.

The cohort was deliberately balanced by Boltz-2's familiarity score and is not a random sample of all Runs N' Poses systems. The calculation uses one seed and one Boltz-2 diffusion sample per system. MSA-1024 is a local memory-constrained protocol, and one long protein failed. Finally, Runs N' Poses does not provide a uniform affinity endpoint for these systems, so none of these structural results establishes affinity-prediction accuracy.

The benchmark survey also changes how I interpret the Week 34 affinity result. The 87 compounds came from only four kinase series, so that calculation is a useful released-model reproduction and a test of within-series ranking, but not a broad test of target-level generalization. The larger public FEP collections provide more diverse follow-up systems, while their prepared structures and congeneric series must remain part of the interpretation.

For next week, I will apply the *Runs N' Poses idea* to affinity benchmarks rather than treating the FEP benchmark names as sufficient evidence of difficulty. Beginning with Nesso-1 and its September 2021 structural cutoff, I will determine how similar each FEP test pocket and ligand is to complexes available before that cutoff, first for FEP+4 and then for selected open Hahn/OpenFF or Ross series. Pocket familiarity, ligand chemical familiarity, and affinity-training overlap will be recorded separately. I will then ask whether Nesso-1's within-series ranking performance changes with familiarity. This is planned work; no such FEP familiarity result is claimed in this report.

References

[1] Peter Škrinjar, Jérôme Eberhardt, Janani Durairaj, and Torsten Schwede. Evaluating generalization in protein–ligand cofolding methods. *Nature Structural & Molecular Biology*, 33:782–794, 2026. doi: 10.1038/s41594-026-01797-5. URL `https://doi.org/10.1038/s41594-026-01797-5`.

[2] Nikhil Shenoy, David Errington, Emmanuel Bengio, Kacper Kapuśniak, Kerstin Klaeser, Yui Tik Pang, Vladimir Radenkovic, Prudencio Tossou, Therence Bois, Andrew Wedlake, and Francesco Di Giovanni. Nesso-1: Accelerating open-source binding affinity predictions. *bioRxiv*, 2026. doi: 10.64898/2026.08.01.742196. URL `https://doi.org/10.64898/2026.08.01.742196`. Preprint, version 1, posted 3 August 2026.

[3] Saro Passaro, Gabriele Corso, Jeremy Wohlwend, Mateo Reveiz, Stephan Thaler, Vignesh Ram Somnath, Noah Getz, Tally Portnoi, Julien Roy, Hannes Stark, David Kwabi-Addo, Dominique Beaini, Tommi Jaakkola, and Regina Barzilay. Boltz-2: Towards accurate and efficient binding affinity prediction. *bioRxiv*, 2025. doi: 10.1101/2025.06.14.659707. URL `https://doi.org/10.1101/2025.06.14.659707`. Preprint, version 1, posted 18 June 2025.

[4] Lingle Wang, Yujie Wu, Yuqing Deng, Byungchan Kim, Levi Pierce, Goran Krilov, Dmitry Lupyan, Shaughnessy Robinson, Markus K. Dahlgren, Jeremy Greenwood, Donna L. Romero, Craig Masse, Jennifer L. Knight, Thomas Steinbrecher, Thijs Beuming, Wolfgang Damm, Edward Harder, Woody Sherman, Mark Brewer, Ronald Wester, Mark Murcko, Leah Frye, Ramy Farid, Teng Lin, David L. Mobley, William L. Jorgensen, Bruce J. Berne, Richard A. Friesner, and Robert Abel. Accurate and reliable prediction of relative ligand binding potency in prospective drug discovery by way of a modern free-energy calculation protocol and force field. *Journal of the American Chemical Society*, 137(7):2695–2703, 2015. doi: 10.1021/ja512751q. URL `https://doi.org/10.1021/ja512751q`.

[5] David F. Hahn et al. Best practices for constructing, preparing, and evaluating protein-ligand binding affinity benchmarks. *Living Journal of Computational Molecular Science*, 4(1):1497, 2022. doi: 10.33011/livecoms.4.1.1497. URL `https://doi.org/10.33011/livecoms.4.1.1497`.

[6] Gregory A. Ross, Chao Lu, Guido Scarabelli, Steven K. Albanese, Evelyne Houang, Robert Abel, Edward D. Harder, and Lingle Wang. The maximal and current accuracy of rigorous protein–ligand binding free energy calculations. *Communications Chemistry*, 6:222, 2023. doi: 10.1038/s42004-023-01019-9. URL `https://doi.org/10.1038/s42004-023-01019-9`.

[7] Open Free Energy Consortium. OpenFE industry benchmark 2024: Public dataset benchmarks. Online documentation, 2024. URL `https://industrybenchmarks2024.readthedocs.io/en/latest/public/overview.html`. Accessed 30 August 2026.

[8] Michael K. Gilson, Jerome Eberhardt, Peter Škrinjar, Janani Durairaj, Xavier Robin, and Andriy Kryshtafovych. Assessment of pharmaceutical protein–ligand pose and affinity predictions

in CASP16. *Proteins: Structure, Function, and Bioinformatics*, 94(1):249–266, 2026. doi: 10.1002/prot.70061. URL https://doi.org/10.1002/prot.70061.

[9] Zied Gaieb, Conor D. Parks, Michael Chiu, Huanwang Yang, Chenghua Shao, W. Patrick Walters, Millard H. Lambert, Neysa Nevins, Scott D. Bembenek, Michael K. Ameriks, Tara Mirzadegan, Stephen K. Burley, Rommie E. Amaro, and Michael K. Gilson. D3R grand challenge 3: Blind prediction of protein–ligand poses and affinity rankings. *Journal of Computer-Aided Molecular Design*, 33(1):1–18, 2019. doi: 10.1007/s10822-018-0180-4. URL https://doi.org/10.1007/s10822-018-0180-4.